Interior smoothness of carrying simplices for competitive population maps

Kaiwen Jin
yui.ui.siki@gmail.com

October 12, 2026

Abstract

A carrying simplex reduces the nonzero long-term dynamics of a competitive population map to an invariant hypersurface. We prove that its positive interior is a C^1 embedded manifold under the standard C^1 per-capita competition and spectral-radius assumptions. Equivalently, its radial function is C^1 in the interior of the probability simplex. The proof uses logarithmic coordinates. Although inverse derivatives lose strict positivity at extinction faces, their weighted columns give a uniform contraction on a fixed cone of positive vectors. The resulting continuous covector annihilates every contingent direction of the logarithmic simplex and forces its Lipschitz graph to be C^1. No boundary smoothness, convexity or permanence assumption is imposed.

1 The regularity question

Consider a discrete population model on $C = [0,\infty)^d$ of the form $F_i(x) = x_i f_i(x)$. The factor f_i describes per-capita growth. Strict decrease of every f_i with respect to every population density expresses both interspecific and intraspecific competition. A carrying simplex is a compact invariant set $\Sigma \subset C \setminus \{0\}$ which meets each nonzero ray once, is unordered for the coordinate order, and carries the asymptotic nonzero dynamics. Unordered means that distinct points cannot satisfy $x_i \le y_i$ for all i. Write

$$\Sigma = \{R(u)u : u \in \Delta\}, \qquad \Delta = \{u \in C : \textstyle\sum_i u_i = 1\}, \tag{1}$$

where R is continuous and positive. This representation gives a geometric reduction by one dimension. Smoothness is needed to use differential geometric methods directly on the reduced dynamics.

Baigent's existence theorem [1, Theorem 1] applies under the hypotheses stated below. The smoothness of the entire positive interior is asked in [1, Section 6]. Mierczyński, Niu and Ruiz-Herrera [2, Section 1, Corollaries 4.8–4.9 and Theorem 4.12] study local linearization and invariant manifolds near interior fixed points. Their local conclusions do not assert smoothness at every interior point. Under additional assumptions, convexity yields a C^1 carrying simplex neatly embedded in the orthant [3, Main Theorem]. That includes a boundary conclusion which is distinct from the question considered here. Logarithmic carrying simplices have also been studied for planar Leslie–Gower and Ricker maps, with conditions ensuring convexity [4].

We answer the interior question for the full following class of maps.

Theorem 1. *Let $d \ge 3$ and $C = [0,\infty)^d$. Let f be C^1 on an open neighborhood of C, and put $F_i(x) = x_i f_i(x)$. Suppose that*

$$f_i(x) > 0, \qquad \partial_j f_i(x) < 0 \quad (x \in C,\ 1 \le i, j \le d).$$

For each i suppose that $q_i e_i$ is a fixed point, with $q_i > 0$. Put $q = (q_1, \dots, q_d)$ and assume

$$\rho(M(x)) < 1 \quad (x \in [0, q]), \qquad M_{ij}(x) = -\frac{x_i}{f_i(x)} \partial_j f_i(x), \qquad [0, q] = \prod_i [0, q_i], \tag{2}$$

where ρ denotes spectral radius. Then the guaranteed carrying simplex has $\Sigma^\circ = \Sigma \cap (0, \infty)^d$ a C^1 embedded $(d-1)$-dimensional submanifold of $(0, \infty)^d$. In (1), $R \in C^1(\operatorname{relint} \Delta)$.

The theorem imposes no assumption on regularity or transverse embedding at extinction faces. In particular, it does not assert neat embedding of the whole simplex. The constants used in its proof may depend on the fixed map and dimension.

The main difficulty is that backward orbits of interior points may approach extinction faces. In physical coordinates some inverse derivative entries then vanish, so a lower bound on every population is unavailable. In logarithmic coordinates the inverse derivatives instead have the form $I + B(x) \operatorname{diag}(x)$, with all entries of B bounded above and below by positive constants. Compactness separates the simplex from the origin, giving a lower bound on the *sum* of the populations. These facts yield uniform contraction of ratios on a fixed positive cone. A second normalization gives a continuous covector whose kernel contains all contingent directions. Unorderedness then turns this constraint into differentiability of the logarithmic graph. The proof below gives both normalizations explicitly.

2 Proof of Theorem 1

All vector inequalities in this section are coordinatewise. Logarithms and exponentials are applied coordinatewise. For a vector r write $\operatorname{osc}(r) = \max_i r_i - \min_i r_i$.

2.1 The logarithmic graph and its inverse derivatives

Each axial fixed point $q_i e_i$ belongs to Σ. Baigent's Theorem 1 [1] gives distance attraction: $\operatorname{dist}(F^n x, \Sigma) \to 0$ for every $x \in C \setminus \{0\}$. For $x = q_i e_i$ this distance is constant, so it is zero; closedness of Σ gives membership. If $x \in \Sigma$ had $x_i > q_i$, it would be comparable with $q_i e_i$. Therefore

$$\Sigma \subset [0, q], \qquad s := \min_{x \in \Sigma} \sum_i x_i > 0, \qquad L := \sum_i q_i. \tag{3}$$

The minimum is positive because Σ is compact and excludes zero.

Under (2) and the sign assumptions, Baigent's Lemma 16(1) [1] states that $F|_{[0,q]}$ is a homeomorphism onto its image. The source hypotheses AS3–AS4 are exactly the positivity, derivative signs and spectral condition here. The axial point on each axis is unique, since $t \mapsto f_i(t e_i)$ is strictly decreasing. Since $F(\Sigma) = \Sigma$, we may set

$$T = (F|_\Sigma)^{-1}.$$

This is a continuous map on the compact physical simplex. Positivity of f_i implies that F preserves support, so T preserves Σ°.

Put $\mathbf{1} = (1, \dots, 1)$ and

$$E = \{w \in \mathbb{R}^d : \textstyle\sum_i w_i = 0\}, \qquad \pi z = z - \frac{1}{d}\Big(\sum_i z_i\Big)\mathbf{1}, \qquad S = \log \Sigma^\circ.$$

The line $w + \mathbb{R}\mathbf{1}$ exponentiates to a positive ray. By the radial representation it meets S exactly once, so there is a function $h : E \to \mathbb{R}$ with

$$S = \{w + h(w)\mathbf{1} : w \in E\}. \tag{4}$$

The exponential preserves order, and hence S is unordered. For $w, v \in E$ this gives

$$\min_i(w_i - v_i + h(w) - h(v)) \le 0 \le \max_i(w_i - v_i + h(w) - h(v)).$$

Consequently

$$|h(w) - h(v)| \le \|w - v\|_\infty \le \|w - v\|_2. \tag{5}$$

We next record bounds for the logarithmic inverse derivative. Define on the closed box

$$K_{ij}(x) = -\frac{\partial_j f_i(x)}{f_i(x)}, \qquad N(x) = K(x)\,\mathrm{diag}(x), \qquad A(x) = (I - N(x))^{-1}. \tag{6}$$

All entries of K are positive. The identity $\det(I - tUV) = \det(I - tVU)$, obtained by the two block eliminations of $\left(\begin{smallmatrix} I & tU \\ V & I \end{smallmatrix}\right)$, shows that N and $M = \mathrm{diag}(x)K$ have identical characteristic polynomials, including at the boundary. Thus $\rho(N) < 1$ and A is continuous on the box. The nonnegative matrix expansion in the proof of [1, Lemma 15] gives $(I-M)^{-1} \ge I$. At a point with all coordinates positive, $N = \mathrm{diag}(x)^{-1} M\,\mathrm{diag}(x)$, so $A \ge I$ there. Approximating a boundary point by $(1-\varepsilon)x + \varepsilon q/2$ and using continuity gives $A \ge I$ everywhere on the box. Set $B = AK$. Then

$$A(x) = I + B(x)\,\mathrm{diag}(x), \qquad 0 < a \le B_{ij}(x) \le b < \infty \quad (x \in [0, q],\ 1 \le i, j \le d). \tag{7}$$

Such a, b exist because $B \ge K > 0$ is continuous on a compact set. Choose

$$\kappa = \frac{a}{2b}, \qquad \eta = \frac{a\kappa s}{1 + bL}, \qquad \tau = 1 - \eta. \tag{8}$$

Here $0 < \kappa \le 1/2$ and $0 < \eta < 1$, so $0 < \tau < 1$.

For the logarithmic map $H(z) = \log F(e^z) = z + \log f(e^z)$ we have $DH(\log x) = I - N(x)$. The inverse function theorem provides a local C^1 inverse at every positive point of Σ. Continuity of T identifies these branches, on nearby points of S, with $G(\log x) = \log Tx$. Thus, for every fixed positive integer n,

$$DG(\log x) = A(Tx), \qquad DG^n(\log x) = P_n(x) := A(T^n x) \cdots A(Tx). \tag{9}$$

In this formula G^n is a local composition along a finite orbit segment.

2.2 Uniform contraction of positive ratios

Consider the cone

$$\mathcal{C}_\kappa = \{v \in (0, \infty)^d : \min_i v_i \ge \kappa \max_i v_i\}.$$

It keeps coordinate ratios bounded. If $v \in \mathcal{C}_\kappa$ and $c = \sum_j x_j v_j$, (7) gives

$$\min_i (A(x)v)_i \ge \min_i v_i + ac, \qquad \max_i (A(x)v)_i \le \max_i v_i + bc.$$

Since $a - \kappa b \ge 0$, $A(x)$ preserves this cone for every point of the box. On positive rays use the Hilbert distance

$$d_H(v, w) = \mathrm{osc}\,((\log(v_i/w_i))_{i=1}^d).$$

It is unchanged by positive scalar multiples. Its diameter on $\mathcal{C}_\kappa$ is at most $D := 2\log(1/\kappa)$.

Lemma 1. *For every $x \in \Sigma$ and $v, w \in \mathcal{C}_\kappa$,*

$$d_H(A(x)v, A(x)w) \le \tau d_H(v, w). \tag{10}$$

Proof. The Jacobian of $r \mapsto \log(A(x)e^r)$ at $e^r = v$ has entries

$$J_{ij} = \frac{A_{ij}(x)v_j}{(A(x)v)_i}.$$

It is row stochastic, meaning that its entries are nonnegative and each row sums to one. From (7) and $v \in \mathcal{C}_\kappa$,

$$J_{ij} \ge \nu_j(x) := \frac{a\kappa x_j}{1 + bL}, \qquad \sum_j \nu_j(x) \ge \eta. \tag{11}$$

Indeed, $A_{ij}(x) \ge a x_j$, $v_j \ge \kappa \max_k v_k$, and $(A(x)v)_i \le (1 + bL) \max_k v_k$. If $m = \sum_j \nu_j(x)$, all rows of $J - \mathbf{1}\nu$ have nonnegative entries and mass $1 - m$. Subtracting the common scalar νr gives

$$\operatorname{osc}(Jr) \le (1 - m) \operatorname{osc}(r) \le \tau \operatorname{osc}(r). \tag{12}$$

Here $m \le a\kappa L/(1 + bL) < 1$.

For v, w in the cone, the segment $r(t) = (1 - t) \log w + t \log v$ has $e^{r(t)} \in \mathcal{C}_\kappa$: every coordinate difference is at least $\log \kappa$. The integral mean-value formula therefore gives

$$\log(A(x)v) - \log(A(x)w) = \int_0^1 J(r(t))(\log v - \log w)\, dt.$$

For each pair of components their difference is bounded by $\tau \operatorname{osc}(\log v - \log w)$, using (12). Taking the largest pairwise difference after integration proves (10). □

Normalize positive vectors by their coordinate sum and write $\widehat{v} = v/(\sum_i v_i)$. The normalized cone section $V = \{v \in \mathcal{C}_\kappa : \sum_i v_i = 1\}$ is compact and nonempty; every coordinate is at least κ/d. For $x \in \Sigma$ let

$$L_n(x) = A(x)A(Fx) \cdots A(F^{n-1}x), \qquad V_n(x) = \{\widehat{L_n(x)v} : v \in V\}.$$

These are nested nonempty compact subsets of V, since every factor preserves the cone. Lemma 1 bounds their Hilbert diameters by $D\tau^n$. Their intersection is consequently a single point $v_*(x)$. To see this in an ordinary norm, if v, w are normalized and $d_H(v, w) \le \delta$, their coordinate ratios straddle one and lie in $[e^{-\delta}, e^{\delta}]$. Hence

$$\|v - w\|_1 \le e^{\delta} - 1. \tag{13}$$

The continuous functions $\widehat{L_n(x)\mathbf{1}}$ converge uniformly to $v_*(x)$ with error at most $e^{D\tau^n} - 1$ in this norm. Thus v_* is continuous on the compact physical Σ, and

$$\sum_i v_{*,i}(x) = 1, \qquad v_*(x) \in \mathcal{C}_\kappa, \qquad A(Tx)v_*(x) = \gamma(x)v_*(Tx), \quad \gamma(x) > 0. \tag{14}$$

The last identity follows from $L_{n+1}(Tx) = A(Tx)L_n(x)$ by normalization and passage to the limit. The fields at boundary points are auxiliary matrix constructions; no logarithm or tangent plane is assigned there.

2.3 A continuous covector from row products

We normalize the inverse derivatives using the positive direction just constructed. For $x \in \Sigma$ set

$$Q_{ij}(x) = \frac{A_{ij}(Tx)v_{*,j}(x)}{(A(Tx)v_*(x))_i}, \qquad Q^{(n)}(x) = Q(T^{n-1}x)\cdots Q(Tx)Q(x). \tag{15}$$

These matrices are row stochastic. Since $v_*(x) \in \mathcal{C}_\kappa$, (11) applies at Tx with input $v_*(x)$: every row of $Q(x)$ dominates $\nu(Tx)$, whose entries sum to at least η. Thus every $Q(x)$ contracts oscillation by $\tau = 1 - \eta$. Therefore

$$\operatorname{osc}(Q^{(n)}(x)r) \le \tau^n \operatorname{osc}(r). \tag{16}$$

The convex hulls of the rows of $Q^{(n)}(x)$ are nested, since the next product is obtained by left multiplication by a row-stochastic matrix. For two rows a_1, a_2 and $\|r\|_\infty \le 1$, (16) gives $|(a_1 - a_2)r| \le 2\tau^n$. Using the elementary duality between the ℓ^1 and ℓ^∞ norms, the entire row hull has ℓ^1 diameter at most $2\tau^n$. It has a unique limiting probability row $p(x)$, and all rows converge uniformly to it. Finite products are continuous, so p is continuous.

Define a row covector by

$$\ell_j(x) = \frac{p_j(x)}{v_{*,j}(x)}. \tag{17}$$

It is continuous and nonnegative, and $\ell(x)v_*(x) = 1$, so $\ell(x)\mathbf{1} > 0$. Individual entries of p need not be positive at the boundary.

For clarity, the product normalization can be written exactly. Put $v_k = v_*(T^k x)$ and $D_k = \operatorname{diag}(v_k)$, and write $A(T^k x)v_{k-1} = \gamma_k v_k$. By (15),

$$Q(T^{k-1}x) = \gamma_k^{-1} D_k^{-1} A(T^k x) D_{k-1}.$$

The successive diagonal factors cancel, giving

$$Q^{(n)}(x) = (\gamma_1 \cdots \gamma_n)^{-1} D_n^{-1} P_n(x) D_0, \qquad P_n(x)v_0 = (\gamma_1 \cdots \gamma_n) v_n.$$

Thus, for every real vector v and every component i,

$$\frac{(P_n(x)v)_i}{(P_n(x)v_*(x))_i} = \left[Q^{(n)}(x) \left(\frac{v_j}{v_{*,j}(x)} \right)_{j=1}^{d} \right]_i \longrightarrow \ell(x)v. \tag{18}$$

All denominators are positive. If $\ell(x)v \neq 0$, then every coordinate of $P_n(x)v$ has that same strict sign for all sufficiently large n.

2.4 Contingent directions and differentiability

For $z \in S$ let $\operatorname{Tan}_S(z)$ consist of limits

$$v = \lim_{k\to\infty} t_k(z_k - z), \qquad z_k \in S, \quad z_k \to z, \quad t_k > 0.$$

Unorderedness excludes strictly positive and strictly negative vectors from this contingent cone. A sequence tending to such a vector would eventually give points strictly comparable with z.

Fix $x \in \Sigma^\circ$ and $v \in \operatorname{Tan}_S(\log x)$. For each fixed n, the local C^1 map G^n from (9) sends nearby points of S into S and hence

$$P_n(x)v \in \operatorname{Tan}_S(\log T^n x).$$

Indeed, its differentiability remainder is $o(\|z_k - \log x\|)$; multiplication by t_k still makes it tend to zero, because $t_k\|z_k - \log x\|$ is bounded whenever the scaled increments have a finite limit. If $\ell(x)v \neq 0$, (18) would make this transported contingent vector strictly signed for some finite n, a contradiction. Consequently

$$\mathrm{Tan}_S(\log x) \subset \ker \ell(x). \tag{19}$$

Only finite local compositions were differentiated in this argument.

Recall that $E = \{w : \sum_i w_i = 0\}$ and $h : E \to \mathbb{R}$ is 1-Lipschitz by (5). At $z = w + h(w)\mathbf{1} = \log x$ define the linear functional on E

$$\Lambda_w(r) = -\frac{\ell(x)r}{\ell(x)\mathbf{1}}. \tag{20}$$

Its denominator is positive. To prove that $Dh(w) = \Lambda_w$, take any sequence $r_k \in E \setminus \{0\}$ tending to zero. By (5), a subsequence has limits

$$\frac{r_k}{\|r_k\|_2} \longrightarrow r_\infty, \qquad \frac{h(w + r_k) - h(w)}{\|r_k\|_2} \longrightarrow c.$$

Then $r_\infty + c\mathbf{1} \in \mathrm{Tan}_S(z)$, so (19) forces $c = \Lambda_w(r_\infty)$. Along this subsequence,

$$\frac{h(w + r_k) - h(w) - \Lambda_w(r_k)}{\|r_k\|_2} \longrightarrow 0.$$

If the full remainder limit were not zero, a sequence staying away from zero would have precisely such a subsequence, giving a contradiction. This proves Fréchet differentiability. The formula (20), continuity of ℓ and of $x = \exp(w + h(w)\mathbf{1})$ show that Dh is continuous. Thus $h \in C^1(E)$.

The graph map $w \mapsto w + h(w)\mathbf{1}$ is injective, has continuous inverse π on its image, and has injective derivative because projecting that derivative gives the identity on E. It is therefore a C^1 embedding. Exponentiation is a C^1 diffeomorphism onto the positive orthant, so Σ° is a C^1 embedded hypersurface.

Finally, for $u \in \operatorname{relint}\Delta$, (4) and the unique intersection with its ray give

$$R(u) = \exp\left(h(\pi \log u) - \frac{1}{d}\sum_i \log u_i\right). \tag{21}$$

This is C^1 on $\operatorname{relint}\Delta$, where all u_i are positive. The map and dimension were arbitrary under the hypotheses of Theorem 1, which completes its proof. □

AI-use statement. OpenAI Codex assisted with literature retrieval, development of the argument, proof checking, and preparation of the manuscript. AI review does not replace independent mathematical review.

References

[1] S. Baigent, *Carrying Simplices for Competitive Maps*, in *Difference Equations, Discrete Dynamical Systems and Applications: ICDEA 2017*, Springer, 2019, pp. 3–29. doi:10.1007/978-3-030-20016-9_1. Author manuscript (Theorem 1, Lemmas 15–16, and Section 6; manuscript pp. 9, 18 and 24).

[2] J. Mierczyński, L. Niu and A. Ruiz-Herrera, *Linearization and invariant manifolds on the carrying simplex for competitive maps*, Journal of Differential Equations **267** (2019), 7385–7410. doi:10.1016/j.jde.2019.08.001. arXiv:1902.08914.

[3] J. Mierczyński, *The C^1 property of convex carrying simplices for competitive maps*, Ergodic Theory and Dynamical Systems **40** (2020), 1335–1350. doi:10.1017/etds.2018.85. arXiv:1801.01032.

[4] H. Naderi Yeganeh and S. Baigent, *Convexity of non-compact carrying simplices in logarithmic coordinates*, Journal of Difference Equations and Applications **30** (2024), 1671–1691. doi:10.1080/10236198.2024.2369215.